

## Lecture 4

# Schur's lemma, orthogonality, and the group algebra

**Objectives.** A single, almost trivial-looking observation about intertwiners — Schur's lemma — turns out to control the entire structure of finite group representations. From it we derive the orthogonality of matrix elements, and, via the regular representation and the group algebra, the two great counting theorems: the squared dimensions of the irreducible representations sum to the group order, and the number of irreducible representations equals the number of conjugacy classes. By the end of the lecture, the representation theory of a finite group is reduced to a finite, sharply constrained bookkeeping problem.

## 4.1 Schur's lemma

Recall from Lecture 3 that an *intertwiner* between representations  $(\rho_V, V)$  and  $(\rho_W, W)$  of  $G$  is a linear map  $T: V \rightarrow W$  with

$$T \rho_V(g) = \rho_W(g) T \quad \text{for all } g \in G;$$

there we required  $T$  invertible and called the representations equivalent. Now we drop invertibility and study the vector space of *all* intertwiners, denoted  $\text{Hom}_G(V, W)$ . The decisive observation is one line long:

**Lemma 4.1.** *If  $T \in \text{Hom}_G(V, W)$ , then  $\ker T$  is an invariant subspace of  $V$  and  $\text{im } T$  is an invariant subspace of  $W$ .*

*Proof.* If  $Tv = 0$  then  $T\rho_V(g)v = \rho_W(g)Tv = 0$ , so  $\rho_V(g)v \in \ker T$ . And  $\rho_W(g)Tv = T\rho_V(g)v \in \text{im } T$ . ■

An *irreducible* representation has no invariant subspaces other than the zero subspace and the whole space. For irreducible representations the lemma therefore leaves only two possibilities.

**Theorem 4.2 (Schur's lemma, first form).** *Let  $V$  and  $W$  be irreducible representations of  $G$  and  $T \in \text{Hom}_G(V, W)$ . Then either  $T = 0$  or  $T$  is an isomorphism (so  $V \cong W$ ). In particular, between inequivalent irreducible representations the only intertwiner is zero.*

*Proof.* Suppose  $T \neq 0$ . By Lemma 4.1,  $\ker T$  is an invariant subspace of the irreducible  $V$ , and  $\ker T \neq V$  since  $T \neq 0$ ; hence  $\ker T = \{0\}$  and  $T$  is injective.

Likewise  $\text{im } T$  is a nonzero invariant subspace of the irreducible  $W$ , hence  $\text{im } T = W$ . So  $T$  is surjective. ■

**Theorem 4.3 (Schur's lemma, second form).** *Let  $(\rho, V)$  be a finite-dimensional complex irreducible representation, and let  $T \in \text{Hom}_G(V, V)$ , i.e.  $T\rho(g) = \rho(g)T$  for all  $g$ . Then  $T = \lambda \text{id}_V$  for some  $\lambda \in \mathbb{C}$ :*

$$\text{Hom}_G(V, V) = \mathbb{C} \text{id}_V.$$

*Proof.* Because  $\mathbb{C}$  is algebraically closed, the characteristic polynomial of  $T$  has a root. So  $T$  has an eigenvalue  $\lambda \in \mathbb{C}$ .

Then  $T - \lambda \text{id}_V$  is again an intertwiner, since sums and scalar multiples of intertwiners intertwine, and  $\text{id}_V$  does trivially. The kernel of this operator is the eigenspace of  $\lambda$ , so it is nonzero.

By Theorem 4.2 a non-invertible intertwiner from  $V$  to  $V$  must vanish. Applied to  $T - \lambda \text{id}_V$ , this gives  $T = \lambda \text{id}_V$ . ■

**Corollary 4.4.** *For complex irreducible representations  $V, W$  of a group  $G$ ,*

$$\dim \text{Hom}_G(V, W) = \begin{cases} 1, & V \cong W, \\ 0, & V \not\cong W. \end{cases}$$

*Proof.* If  $V \not\cong W$  this is the first form. If  $V \cong W$ , fix an isomorphism  $T_0 \in \text{Hom}_G(V, W)$ ; for any  $T \in \text{Hom}_G(V, W)$  the composite  $T_0^{-1}T \in \text{Hom}_G(V, V)$  is a scalar  $\lambda \text{id}$  by the second form, so  $T = \lambda T_0$ . ■

**Corollary 4.5 (Abelian groups).** *Every complex irreducible representation of an abelian group is one-dimensional.*

*Proof.* Let  $(\rho, V)$  be irreducible and  $G$  abelian. For fixed  $h \in G$  the operator  $\rho(h)$  commutes with every  $\rho(g)$  — that is,  $\rho(h) \in \text{Hom}_G(V, V)$  — so  $\rho(h) = \lambda_h \text{id}_V$  by the second form.

So every group element acts as a scalar. Then every subspace of  $V$  is invariant, and irreducibility forces  $\dim V = 1$ . ■

Schur's lemma explains what Lecture 3 found by hand: the rotation representation of  $C_3$  had to split into one-dimensional pieces over  $\mathbb{C}$ . The one-dimensional representations  $\chi_k(r^m) = e^{2\pi i km/n}$  of  $C_n$  are therefore the only candidates for its irreducible representations. That they form the *complete* list is shown in Section 4.3.

**Remark 4.6 (Complex numbers are essential).** The second form fails over  $\mathbb{R}$ , where eigenvalues need not exist. Let  $C_4 = \langle r \rangle$  act irreducibly on  $\mathbb{R}^2$  by the quarter-turn  $J = R(\pi/2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . Every matrix  $a\mathbb{1} + bJ$  commutes with the action. So the *commutant*, the space of all matrices commuting with the action, is two-dimensional. It is a copy of  $\mathbb{C}$  among the real  $2 \times 2$  matrices, with  $J$  playing the role of  $i$ .

Over  $\mathbb{C}$  the representation splits into  $\chi_1 \oplus \chi_{-1}$ , and Schur's lemma holds again. Throughout this course “irreducible” means irreducible over  $\mathbb{C}$  unless flagged otherwise.

**Remark 4.7 (The physics of Schur's lemma).** Read physically, the second form says: *an operator that commutes with an irreducible symmetry action is a multiple of the identity*. If the Hamiltonian of a system commutes with all the  $\rho(g)$  — the defining property of a symmetry, as Lecture 7 will set up — then on an irreducible representation occurring with multiplicity one the Hamiltonian is forced to be a scalar: one energy for the whole representation.

More generally, on the sum of all  $m_i$  copies of  $V^{(i)}$  — which Lecture 5 will name the *isotypic component* and write  $V^{(i)} \otimes \mathbb{C}^{m_i}$  — the symmetry acts as  $\rho^{(i)}(g) \otimes \mathbb{1}$ , and its commuting operators have the form  $\mathbb{1} \otimes F_i$ . They may therefore mix the  $m_i$  equivalent copies;

after diagonalising  $F_i$ , each of its eigenvalues still has the  $\dim V^{(i)}$ -fold multiplet forced by symmetry.

More precisely, if an eigenvalue of  $F_i$  has multiplicity  $q$ , its total degeneracy on the isotypic component is  $q \dim V^{(i)}$ ; only the factor  $\dim V^{(i)}$  is forced by this complex irreducible type. Irreducible representations are thus the natural candidates for the degenerate multiplets of quantum mechanics, and Schur's lemma is the reason. The same mechanism, applied to operators that intertwine *different* irreducible blocks, will give the Wigner–Eckart theorem and the selection rules of Lecture 7.

## 4.2 Orthogonality of matrix elements

Schur's lemma is a statement about operators. Combined with the averaging trick of Lecture 3, it becomes a statement about *numbers* — the matrix elements of the irreducible representations — and these numbers turn out to be orthogonal in a precise sense.

Fix once and for all a complete list  $\rho^{(1)}, \rho^{(2)}, \dots$  of pairwise inequivalent irreducible representations of the finite group  $G$ , on spaces  $V^{(i)}$  of dimensions  $n_i$ , and choose bases making the matrices  $D^{(i)}(g)$  *unitary* (always possible, Lecture 3). The completeness of the list — that there are finitely many and that we have them all — is established in the next section; nothing in the present one depends on it.

We regard each matrix element as a complex-valued *function on the group*,  $g \mapsto D_{\mu\nu}^{(i)}(g)$ , and we equip the  $|G|$ -dimensional space of all functions  $\psi: G \rightarrow \mathbb{C}$  with the inner product

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\varphi(g)} \psi(g),$$

conjugate-linear in the first argument as always.

The step from Schur's lemma to orthogonality goes through the following averaged intertwiner.

**Lemma 4.8.** *For any linear map  $A: V^{(j)} \rightarrow V^{(i)}$ , the average*

$$T_A = \frac{1}{|G|} \sum_{g \in G} \rho^{(i)}(g) A \rho^{(j)}(g)^{-1}$$

*is an intertwiner:  $T_A \in \text{Hom}_G(V^{(j)}, V^{(i)})$ .*

*Proof.* For  $h \in G$ ,

$$\rho^{(i)}(h) T_A \rho^{(j)}(h)^{-1} = \frac{1}{|G|} \sum_{g \in G} \rho^{(i)}(hg) A \rho^{(j)}(hg)^{-1} = T_A,$$

since  $g \mapsto hg$  merely reorders the sum (Latin-square property, Lecture 1). Multiply on the right by  $\rho^{(j)}(h)$ . ■

**Theorem 4.9 (Great orthogonality theorem).** *With unitary matrices as above,*

$$\left\langle D_{\mu\nu}^{(i)}, D_{\alpha\beta}^{(j)} \right\rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{D_{\mu\nu}^{(i)}(g)} D_{\alpha\beta}^{(j)}(g) = \frac{1}{n_i} \delta_{ij} \delta_{\mu\alpha} \delta_{\nu\beta}.$$

*Equivalently: the rescaled matrix elements  $\sqrt{n_i} D_{\mu\nu}^{(i)}$  form an orthonormal set of functions on  $G$ .*

*Proof.* Apply Lemma 4.8 with  $A = E^{\nu\beta}$ , the matrix with a single 1 in row  $\nu$ , column  $\beta$ , and zeros elsewhere. In matrix form, the  $(\mu, \alpha)$  entry of  $T_A$  is

$$(T_{E^{\nu\beta}})_{\mu\alpha} = \frac{1}{|G|} \sum_g D_{\mu\nu}^{(i)}(g) D_{\beta\alpha}^{(j)}(g^{-1}) = \frac{1}{|G|} \sum_g D_{\mu\nu}^{(i)}(g) \overline{D_{\alpha\beta}^{(j)}(g)}.$$

Unitarity was used in the last step. It gives  $D^{(j)}(g^{-1}) = D^{(j)}(g)^\dagger$ , so  $D_{\beta\alpha}^{(j)}(g^{-1}) = \overline{D_{\alpha\beta}^{(j)}(g)}$ .

Case  $i \neq j$ . The representations are inequivalent, so Theorem 4.2 gives  $T_A = 0$  for every  $A$ : each entry above vanishes, which is the assertion with  $\delta_{ij} = 0$ .

Case  $i = j$ . By Theorem 4.3,  $T_A = \lambda(A) \text{id}$ , and taking traces fixes the scalar:

$$\lambda(A) n_i = \text{tr } T_A = \frac{1}{|G|} \sum_g \text{tr}(\rho^{(i)}(g) A \rho^{(i)}(g)^{-1}) = \text{tr } A,$$

so  $\lambda(E^{\nu\beta}) = \delta_{\nu\beta}/n_i$ . The  $(\mu, \alpha)$  entry of  $\lambda \text{id}$  is  $\delta_{\nu\beta} \delta_{\mu\alpha}/n_i$ .

Combine the two cases and take the complex conjugate of the whole identity. This changes nothing, since the right-hand side is real, and gives the stated formula. ■

**Corollary 4.10.**  $\sum_i n_i^2 \leq |G|$ . Hence a finite group has finitely many inequivalent irreducible representations, and their squared dimensions sum to at most the group order.

*Proof.* First, finiteness. Suppose there were more than  $|G|$  inequivalent irreducibles; choose  $|G| + 1$  of them and one matrix element from each. By the great orthogonality theorem these are nonzero and mutually orthogonal in the  $|G|$ -dimensional space of functions on  $G$ , which is impossible. So the list is finite, and the sum  $\sum_i n_i^2$  has finitely many terms.

Now the bound. The functions  $\sqrt{n_i} D_{\mu\nu}^{(i)}$ , of which there are  $\sum_i n_i^2$ , are orthonormal by the same theorem, hence linearly independent, in that same  $|G|$ -dimensional space. So there cannot be more of them than the dimension of that space. ■

The inequality will sharpen to an equality in Section 4.3. First, let us check the theorem on a concrete example.

#### Tutorial

**Example 4.11 (Orthogonality in  $D_3$ , numerically).** The group  $D_3$  has, among its irreducible representations, the trivial representation  $\mathbf{1}$ , the orientation representation  $\nu$  (both one-dimensional), and the two-dimensional defining representation with the unitary (indeed real orthogonal) matrices of Lecture 3, namely  $D(r^k) = R(2\pi k/3)$  and  $D(r^k s) = R(2\pi k/3) \text{diag}(-1, 1)$ . Writing  $c_k = \cos \frac{2\pi k}{3}$ ,  $s_k = \sin \frac{2\pi k}{3}$ , the matrix-element functions take the values

| $g$         | $e$ | $r$                   | $r^2$                | $s$ | $rs$                  | $r^2 s$              |
|-------------|-----|-----------------------|----------------------|-----|-----------------------|----------------------|
| $\nu(g)$    | 1   | 1                     | 1                    | -1  | -1                    | -1                   |
| $D_{11}(g)$ | 1   | $-\frac{1}{2}$        | $-\frac{1}{2}$       | -1  | $\frac{1}{2}$         | $\frac{1}{2}$        |
| $D_{12}(g)$ | 0   | $-\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{3}}{2}$ | 0   | $-\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{3}}{2}$ |

(the reflection columns use  $D(r^k s)_{11} = -c_k$ ,  $D(r^k s)_{12} = -s_k$ ). Then, entry by entry:

$$\langle D_{11}, D_{11} \rangle_G = \frac{1}{6} \left( 1 + \frac{1}{4} + \frac{1}{4} + 1 + \frac{1}{4} + \frac{1}{4} \right) = \frac{1}{2} = \frac{1}{n_i}, \quad \langle D_{12}, D_{12} \rangle_G = \frac{1}{6} \cdot 3 = \frac{1}{2},$$

while the cross terms cancel in pairs:

$$\langle D_{11}, D_{12} \rangle_G = 0, \quad \langle \mathbf{1}, \nu \rangle_G = \frac{1}{6}(1 + 1 + 1 - 1 - 1 - 1) = 0, \quad \langle \nu, D_{11} \rangle_G = 0.$$

Six functions  $(1 + 1 + 4)$ . The five sums displayed are only a sample. The great orthogonality theorem, proved above, guarantees that all six are pairwise orthogonal in the six-dimensional function space. So for  $D_3$  the matrix elements fill the whole space. This is a first instance of the completeness theorem proved later.

### 4.3 The group algebra and the regular representation

To convert the inequality of Corollary 4.10 into an equality, we use one special representation that contains every irreducible representation: the group acting on itself.

**Definition 4.12 (Group algebra).** The group algebra  $\mathbb{C}[G]$  is the  $|G|$ -dimensional vector space with one basis vector  $e_g$  for each  $g \in G$ , equipped with the bilinear multiplication extending the group law:

$$\left( \sum_g a_g e_g \right) \left( \sum_h b_h e_h \right) = \sum_{g,h} a_g b_h e_{gh}.$$

It is an associative algebra (a vector space with a bilinear associative product) with unit  $e_e$ , in which  $G$  sits as the basis. The regular representation of  $G$  is left multiplication on this space:

$$\rho^{\text{reg}}(g) e_h = e_{gh},$$

the permutation representation (Lecture 3) of the action of  $G$  on itself by left translation; its dimension is  $|G|$ .

Every representation  $\rho$  of the group extends to the algebra by linearity,  $\rho(\sum a_g e_g) := \sum a_g \rho(g)$ , a fact we will use silently below and seriously in the starred subsection.

Two short lemmas reveal the structure of the regular representation. The first says that an intertwiner out of  $\mathbb{C}[G]$  is freely specified by one vector; the second says that intertwiner dimensions count multiplicities.

**Lemma 4.13.** For any representation  $(\rho_W, W)$  of  $G$ , the evaluation map

$$\text{Hom}_G(\mathbb{C}[G], W) \longrightarrow W, \quad T \longmapsto T(e_e),$$

is a linear isomorphism. In particular  $\dim \text{Hom}_G(\mathbb{C}[G], W) = \dim W$ .

*Proof.* An intertwiner is determined by its value at  $e_e$ :

$$T(e_g) = T(\rho^{\text{reg}}(g)e_e) = \rho_W(g)T(e_e).$$

Conversely, given any  $w \in W$ , define  $T(e_g) := \rho_W(g)w$  and extend linearly; then  $T(\rho^{\text{reg}}(h)e_g) = T(e_{hg}) = \rho_W(hg)w = \rho_W(h)T(e_g)$ , so  $T$  intertwines. The two constructions are mutually inverse and linear in  $w$ . ■

**Lemma 4.14 (Multiplicities count intertwiners).** Let  $V \cong \bigoplus_i m_i V^{(i)}$  be a decomposition of a representation into pairwise inequivalent irreducibles  $V^{(i)}$  with multiplicities  $m_i$ . Then for each  $j$ ,

$$\dim \text{Hom}_G(V, V^{(j)}) = m_j = \dim \text{Hom}_G(V^{(j)}, V).$$

*Proof.*  $\text{Hom}_G$  is additive in direct sums in either slot. An intertwiner from a direct sum restricts to an intertwiner on each invariant summand (since the restriction still commutes with  $G$ ), and it is determined by these restrictions. Conversely, any collection of such intertwiners assembles into one. Intertwiners into a direct sum are handled the same way, by composing with the projections.

Hence

$$\dim \text{Hom}_G\left(\bigoplus_i m_i V^{(i)}, V^{(j)}\right) = \sum_i m_i \dim \text{Hom}_G(V^{(i)}, V^{(j)}) = \sum_i m_i \delta_{ij} = m_j,$$

by Corollary 4.4. The second equality is proved in the same way. ■

Lemma 4.14 answers a question left open in Lecture 3. The multiplicities in a complete decomposition are the dimensions of intertwiner spaces. These spaces are defined without reference to any decomposition. So *the multiplicities are unique*, whatever basis or splitting is used to compute them.

**Theorem 4.15 (Decomposition of the regular representation).** *Every irreducible representation  $V^{(i)}$  of the finite group  $G$  occurs in the regular representation, with multiplicity equal to its dimension:*

$$\mathbb{C}[G] \cong \bigoplus_i n_i V^{(i)}, \quad n_i = \dim V^{(i)},$$

the sum running over all inequivalent irreducible representations. Consequently

$$\sum_i n_i^2 = |G|$$

*Proof.* By Maschke's theorem,  $\mathbb{C}[G] \cong \bigoplus_i m_i V^{(i)}$  for some multiplicities  $m_i \geq 0$ . A priori, some irreducibles might be absent. By Lemmas 4.13 and 4.14,

$$m_i = \dim \text{Hom}_G(\mathbb{C}[G], V^{(i)}) = \dim V^{(i)} = n_i > 0 :$$

every irreducible occurs,  $n_i$  times. Comparing dimensions,  $|G| = \dim \mathbb{C}[G] = \sum_i n_i \cdot n_i$ . ■

The regular representation thus contains every irreducible representation. Decompose it, and each irreducible appears as many times as its dimension. This counting identity strongly constrains the possible dimensions.

**Example 4.16.**

- (i) *Cyclic groups, completeness at last.* The  $n$  representations  $\chi_0, \dots, \chi_{n-1}$  of  $C_n$  from Lecture 3 satisfy  $\sum 1^2 = n = |C_n|$ . So they are all the irreducible representations of  $C_n$ , as promised.
- (ii)  *$S_3$  is finished too.* The trivial, sign and standard representations give  $1^2 + 1^2 + 2^2 = 6 = |S_3|$ . So the list found in Lecture 3 is complete, and no further search is needed.
- (iii) *Constraint solving.* Take any nonabelian group of order 8. We look for dimension patterns  $(n_1, n_2, \dots)$  satisfying three conditions:  $\sum n_i^2 = 8$ ,  $n_1 = 1$  (the trivial representation always exists), and not all  $n_i = 1$  (otherwise, by Corollary 4.22 below, the group would be abelian). Exactly one pattern remains:

$$8 = 1 + 1 + 1 + 1 + 4 :$$

four one-dimensional representations and one two-dimensional. Both nonabelian groups of order 8, the dihedral group  $D_4$  and the quaternion group  $Q_8$  of Sheet 4, must have these dimensions before we compute anything.

## 4.4 Counting the irreducibles: conjugacy classes

How many irreducible representations does a finite group have? The answer ties representation theory back to the conjugacy classes of Lecture 2. Together with  $\sum n_i^2 = |G|$ , it is the second result on which the character tables of Lecture 5 rely.

**Theorem 4.17.** *The number of inequivalent complex irreducible representations of a finite group equals the number of its conjugacy classes.*

The proof identifies both numbers with the dimension of one and the same space: the centre of the group algebra,

$$Z(\mathbb{C}[G]) = \{ z \in \mathbb{C}[G] : za = az \text{ for all } a \in \mathbb{C}[G] \}.$$

That the dimension of the centre equals the class count is elementary, and we prove it at once. That it also equals the representation count is proved in the starred material below.

**Lemma 4.18 (Class sums).** *For each conjugacy class  $C \subseteq G$  let  $z_C = \sum_{g \in C} e_g \in \mathbb{C}[G]$ . The class sums form a basis of  $Z(\mathbb{C}[G])$ ; in particular*

$$\dim Z(\mathbb{C}[G]) = \#\{\text{conjugacy classes of } G\}.$$

*Proof.* An element  $a = \sum_g a_g e_g$  is central iff it commutes with every basis element, i.e.  $e_h a e_h^{-1} = a$  for all  $h$ . The left side is  $\sum_g a_g e_{hgh^{-1}} = \sum_g a_{h^{-1}gh} e_g$ . So an element is central exactly when its coefficient function is constant on each conjugacy class, i.e.  $a_{hgh^{-1}} = a_g$  for all  $g, h$ .

Such elements are precisely the linear combinations of class sums, and the  $z_C$  are linearly independent because distinct classes are disjoint. ■

*Structure of the remaining argument.* Extending each irreducible representation linearly to the algebra gives a map

$$\Phi: \mathbb{C}[G] \longrightarrow \bigoplus_i \text{End}(V^{(i)}), \quad \Phi(a) = (\rho^{(1)}(a), \rho^{(2)}(a), \dots),$$

into the direct sum of full matrix algebras. We show that  $\Phi$  is an *isomorphism of algebras*. It is injective because the regular representation is faithful and built from the  $V^{(i)}$ . It is surjective by the dimension count  $\sum n_i^2 = |G|$  of Theorem 4.15.

The centre of a direct sum of matrix algebras is easy to compute. The centre of each block consists of scalar matrices only (Schur's lemma again, in the language of algebras), so each block contributes one independent scalar. Hence  $\dim Z(\mathbb{C}[G])$  equals the number of blocks, i.e. of irreducible representations, and Lemma 4.18 finishes the proof. The details follow, starred; the statement of Theorem 4.17 is what the rest of the course uses.

Reading

### The group algebra is a sum of matrix algebras

**Theorem 4.19 (★).** *The map  $\Phi$  above is an isomorphism of algebras:*

$$\mathbb{C}[G] \cong \bigoplus_i \text{End}(V^{(i)}) \cong \bigoplus_i M_{n_i}(\mathbb{C}).$$

*Proof (★).* The map  $\Phi$  is an algebra homomorphism since each extended  $\rho^{(i)}$  is: it is linear and preserves products of basis elements.

*Injectivity.* Suppose  $\Phi(a) = 0$ , i.e.  $a$  acts as zero in every irreducible representation. By Maschke's theorem every representation is a direct sum of irreducibles, so  $a$  acts as zero in every representation of  $G$  — in particular in the regular representation. But the regular representation detects every algebra element:  $\rho^{\text{reg}}(a) e_e = a \cdot e_e = a e_e$ . Hence  $a = 0$ .

*Surjectivity.* By Theorem 4.15 both sides have the same finite dimension,  $\dim \mathbb{C}[G] = |G| = \sum_i n_i^2 = \sum_i \dim \text{End}(V^{(i)})$ . An injective linear map between spaces of equal finite dimension is surjective. ■

**Lemma 4.20 (★).** *The centre of the matrix algebra  $\text{End}(V) \cong M_n(\mathbb{C})$  consists of the scalar matrices:  $Z(M_n(\mathbb{C})) = \mathbb{C} \mathbf{1}$ .*

*Proof (★).* Let  $Z$  commute with all matrices, in particular with the matrix units  $E^{\mu\nu}$ . Comparing entries of  $ZE^{\mu\nu} = E^{\mu\nu}Z$  gives  $Z_{\alpha\mu}\delta_{\nu\beta} = \delta_{\alpha\mu}Z_{\nu\beta}$  for all indices; setting  $\alpha = \mu$ ,  $\beta = \nu$  yields  $Z_{\mu\mu} = Z_{\nu\nu}$ . Setting  $\beta = \nu$ ,  $\alpha \neq \mu$  yields  $Z_{\alpha\mu} = 0$ , so the off-diagonal entries vanish. ■

*Proof of Theorem 4.17 (★).* An element of the centre of a direct sum of algebras is a tuple of central elements of the summands, so by Theorem 4.19 and Lemma 4.20,

$$Z(\mathbb{C}[G]) \cong \bigoplus_i Z(\text{End}(V^{(i)})) = \bigoplus_i \mathbb{C} \mathbf{1}_{V^{(i)}},$$

of dimension equal to the number of irreducible representations. By Lemma 4.18 the same dimension is the number of conjugacy classes. ■

Theorem 4.19 matters beyond this proof. It says that the group algebra of a finite group is, as an algebra, just a direct sum of full matrix algebras. The block sizes are the dimensions of the irreducible representations. The great orthogonality theorem is this statement seen through the inner product; the character theory of Lecture 5 is this statement seen through the trace.

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With both counting theorems available, representation theory becomes arithmetic.

**Example 4.21 (Dimension bookkeeping).**

| $G$             | #classes | $ G $ | dimensions $(n_i)$ |
|-----------------|----------|-------|--------------------|
| $C_n$           | $n$      | $n$   | $(1, 1, \dots, 1)$ |
| $S_3 \cong D_3$ | 3        | 6     | $(1, 1, 2)$        |
| $D_4$           | 5        | 8     | $(1, 1, 1, 1, 2)$  |
| $Q_8$           | 5        | 8     | $(1, 1, 1, 1, 2)$  |
| $D_5$           | 4        | 10    | $(1, 1, 2, 2)$     |

In each line the dimensions are *forced*. They are the unique multiset of positive integers with as many entries as there are classes, one of them 1 (the trivial representation), and with squares summing to  $|G|$ . The class counts of  $D_4$  and  $D_5$  come from Lecture 2. The quaternion group  $Q_8$  of Sheet 4 has five classes,  $\{1\}$ ,  $\{-1\}$ ,  $\{\pm i\}$ ,  $\{\pm j\}$  and  $\{\pm k\}$ . The element  $-1$  is central, and conjugation by  $j$  gives  $jij^{-1} = -i$ ; the other pairs work the same way.

The table also carries a warning:  $D_4$  and  $Q_8$  have identical dimension data, yet are not isomorphic. Representation dimensions do not determine the group.

**Corollary 4.22.** *A finite group is abelian if and only if all of its irreducible representations are one-dimensional, and in that case it has exactly  $|G|$  of them.*

*Proof.* If  $G$  is abelian, Corollary 4.5 gives  $n_i = 1$  for all  $i$ , and then  $\sum_i n_i^2 = |G|$  forces  $|G|$  of them.

Conversely, suppose all  $n_i = 1$ . Then the number of irreducibles is  $\sum n_i^2 = |G|$ , which by Theorem 4.17 is the number of conjugacy classes. So every class is a singleton, i.e.  $ghg^{-1} = g$  for all  $g, h$ , and the group is abelian. ■

Reading

## 4.5 Fourier analysis on a finite group

The two counting theorems turn the orthogonality of matrix elements into a completeness statement.

**Theorem 4.23 (Completeness of matrix elements).** *The functions  $\sqrt{n_i} D_{\mu\nu}^{(i)}$  ( $i$  over the irreducible representations,  $1 \leq \mu, \nu \leq n_i$ , matrices unitary) form an orthonormal basis of the space of all functions  $G \rightarrow \mathbb{C}$  with the inner product  $\langle \cdot, \cdot \rangle_G$ . Every function  $\psi$  on  $G$  expands uniquely as*

$$\psi(g) = \sum_i \sum_{\mu, \nu=1}^{n_i} c_{\mu\nu}^{(i)} D_{\mu\nu}^{(i)}(g), \quad c_{\mu\nu}^{(i)} = n_i \left\langle D_{\mu\nu}^{(i)}, \psi \right\rangle_G.$$

*Proof.* Orthonormality is Theorem 4.9; the count  $\sum_i n_i^2 = |G|$  (Theorem 4.15) says the set has exactly the dimension of the function space, so it is a basis. The coefficient formula is the standard orthonormal expansion. ■

For  $G = C_n$  the matrix elements are the one-dimensional representations  $\chi_k(r^m) = e^{2\pi i k m / n}$ . Lecture 5 will call them characters. Here the theorem is the discrete Fourier transform. An arbitrary function of  $m$  expands in exponential modes, with the familiar coefficient integrals replaced by averages over the group.

Seen from this angle, the whole lecture generalises Fourier analysis to noncommutative groups, where the “frequencies” become matrices. The same idea, carried over from finite groups to compact Lie groups in Part II, will exhibit the spherical harmonics as matrix elements of the rotation group, with the Peter–Weyl theorem in the role of Theorem 4.23.

### Summary.

- *Schur's lemma.* An intertwiner of irreducibles is zero or an isomorphism (Theorem 4.2); over  $\mathbb{C}$  it is a scalar (Theorem 4.3), so abelian groups have one-dimensional irreducibles (Corollary 4.5).
- *Great orthogonality theorem.* Matrix elements of the irreducibles are mutually orthogonal, with norm fixed by the dimension (Theorem 4.9).
- *Sum of squares.*  $\sum_i n_i^2 = |G|$ . This follows from decomposing the regular representation (Theorem 4.15).
- *Counting.* The number of irreducibles equals the number of conjugacy classes (Theorem 4.17).
- *Completeness.* The reading shows that the matrix elements of the irreducibles form a basis of the functions on  $G$ , and reads the expansion as a Fourier transform.

**Tutorial.** Exercise Sheet 4.

## References

Serre, *Linear Representations of Finite Groups*, §§2.2–2.5 (Schur's lemma and the orthogonality relations); Jones §§4.1–4.3; Tung §§3.4–3.5, 3.7; Fulton–Harris §§1.2, 2.2, 3.4.